

WardSense

Federated Edge AI for early deterioration monitoring

Warning nurses hours before deterioration becomes obvious — without a single byte of patient data leaving the hospital.

THE PROBLEM

Deterioration is visible in the data hours before anyone sees it

1 Nobody is watching between rounds Wards check observations every 4–6 hours. Patients deteriorate continuously. The warning signs sit unread in a chart.	2 The obvious fix is illegal Streaming vitals to a cloud AI service is exactly what patient-privacy regulation forbids. Data cannot leave the hospital.	3 Alerts without context get ignored A red number tells a nurse something changed. It does not tell them why it matters for this patient.
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BETWEEN
OBSERVATIONS

4–6 hours unmonitored

WardSense forecasts the same window — continuously, on-device.

WHAT WE BUILT

Two systems, one junction

The detector

Reads 12 hours of six vitals, forecasts deterioration 4–6 hours ahead.

Under 100K parameters. Trained across hospitals without sharing data.

The assistant

A local language model answering questions from that patient's own record.

Runs offline. Strict per-patient isolation, adversarially tested.

They meet at exactly one point — and that is the contribution

When the detector raises an alert, the abnormal vitals themselves become the search query — pulling the history that explains this specific deterioration, then writing it in plain English. The output of a privacy-protected federated model becomes the semantic input to a grounded language model.

THE FLOW

How one alert becomes a clinical summary

1	2	3	4	5
Vitals arrive Heart rate, BP, oxygen, respiration, temperature, glucose — every hour	Scored on-device 12-hour window through the detector. No network round trip	Risk crosses threshold NEWS2 ≥ 5 — the NHS escalation point. Alert raised	Record retrieved The abnormal vitals become the search query for this patient's history	Nurse reads why Plain-English summary: what is abnormal, and what explains it
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Every step happens on hardware the hospital owns. If the internet fails, monitoring continues.

EDGE AI — MEASURED, NOT ESTIMATED

The whole stack fits in 6 GB

We built to a hard constraint: a consumer laptop GPU. The resulting envelope is the evidence — a stack that fits commodity hardware runs on a ward mini-PC; one that needs a datacenter GPU is not deployable at all.

6 GB total VRAM budget NVIDIA RTX 4050	0.03 GB detector footprint leaves the GPU free	66 tok/s local language model fully GPU-resident	0 external API calls at any point
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What building under a real budget taught us

Differential privacy costs roughly 5.7× more compute on recurrent models than convolutional ones — a trade-off that only surfaces when you measure on the target hardware, and one that directly decides which architecture is viable at the edge.

PRIVACY BY CONSTRUCTION

The model travels. The data never does.

Federated learning Each hospital trains on its own patients and sends back only what the model learned — never a patient record.	Differential privacy Calibrated noise is added on-device before any update leaves, so individuals cannot be reconstructed from the gradients.	Provable budget Privacy loss is accounted across every training round, not per round — a mistake that silently under-reports by 2.4x. MEASURED PRIVACY BUDGET $\epsilon \approx 2.0$ at $\delta = 10^{-5}$, tracked across the full training run using Rényi accounting
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Federated learning alone still leaks — gradients can be partly inverted. Differential privacy is the mathematical guarantee that closes it.

THE CONTRIBUTION

An alert a nurse can act on

WHAT MOST SYSTEMS SHOW

Bed 7 — RISK 0.87

A number changed. The nurse must work out what it means, for which patient, and why — during a shift where twelve other things are also urgent.

Alert fatigue. Ignored.

WHAT WARDSense SHOWS

Bed 7 — deteriorating, NEWS2 9

Respiratory rate 29 (normal 12–20), oxygen 90% (normal ≥ 96), blood pressure fallen to 93.

Known COPD, managed with an inhaler; ex-smoker. Post-operative day 2, reported breathlessness on exertion yesterday.

Escalation threshold crossed four hours ago and sustained.

Context. Actionable.

NOT A SIMULATION

Real patients. Real hospitals. Already tested.

40,336 de-identified ICU patients	2 independent hospital systems	~800k hourly training windows	4–6 h forecast horizon, ahead of onset
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Labels a clinician recognises

We score every hour with NEWS2 — the early-warning chart used across NHS hospitals — rather than inventing thresholds. Our labels were cross-checked against clinically adjudicated outcomes in the same records.

Sites differ in what they measure

One hospital records blood glucose almost twice as often as the other. That mismatch is exactly the federation problem real deployments hit — and it cannot be reproduced with synthetic data.

EVERY LAYER RUNS LOCALLY

Technology

LAYER	STACK	DETAIL
Detector	PyTorch sequence models · DPLSTM / DPGRU, CNN baseline	< 100K parameters
Federation	Flower, containers over gRPC	genuine multi-site
Privacy	Opacus DP-SGD · Rényi accounting	$\epsilon \approx 2.0$ measured
Language model	Llama 3.2 3B, 4-bit · served by Ollama	66 tok/s, offline
Retrieval	ChromaDB, local embeddings · per-patient isolation	no external calls
Platform	FastAPI · Streamlit · SQLite + AES-256	encrypted at rest

Where we are

BUILT AND MEASURED	NEXT
✓ Dataset secured, tested and processed	→ Full privacy-versus-accuracy experiment sweep
✓ Privacy pipeline verified end to end at $\epsilon \approx 2.0$	→ Alert-to-summary bridge and clinician dashboard
✓ Federated averaging converging across sites	→ Three-layer evaluation of generated summaries
✓ Local language model benchmarked on-device	→ Voice input for bedside accessibility
✓ Compute budget measured on target hardware	→ Split across dedicated edge hardware

Privacy is the constraint. The edge is the answer.

A deterioration monitor that learns from every hospital, and reveals nothing about any patient — running inside 6 GB on hardware a ward can afford.

6 GB

full stack

£ ≈ 2.0

privacy budget

40,336

real patients

4–6 h

early warning